

Data Analytics – Stage Two (2A)

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Title: HNG Ride SQL Business Analysis

Introduction

This report presents a comprehensive analysis of the HNG ride-sharing platform's operational and financial performance, offering key insights into driver efficiency, rider behavior, revenue growth, and overall business health. The findings highlight strong performance metrics among top-tier drivers, consistent quarter-over-quarter revenue expansion, and encouraging indicators of customer retention. Notably, high performing drivers continue to uphold excellent service standards, while sustained rider engagement strategies contribute significantly to long-term customer loyalty and growth.

The analysis was conducted using SQL queries on the platform's operational database, examining data across riders, drivers, rides, and payments tables. All queries focused on completed rides with successful payments to ensure data quality and business relevance. The timeframe covers from June 2021 – December 2024.

Business Analysis

1. Top 10 Longest Completed Rides (by distance)

Objective: To find the top 10 longest rides (by distance), including driver name, rider name, pickup/drop-off cities, and payment method.

```
select
    t1.distance_km,
    t2.name as driver_name,
    t3.name as rider_name,
    t1.pickup_city,
    t1.dropoff_city,
    t4.method as payment_method
from
    rides t1
join
    drivers t2 on t1.driver_id = t2.driver_id
join
    riders t3 on t1.rider_id = t3.rider_id
join
    payments t4 on t1.ride_id = t4.ride_id
where
    t4.amount > 0
order by
    t1.distance_km desc
limit 10;
```

Fig1.

distance_km	driver_name	rider_name	pickup_city	dropoff_city	payment_method
30	Driver_1774	Rider_292	Calgary	Los Angeles	voucher
30	Driver_1886	Rider_6131	Ottawa	Vancouver	voucher
29.99	Driver_932	Rider_6330	Chicago	Boston	card
29.99	Driver_1406	Rider_3294	New York	Toronto	cash
29.99	Driver_1093	Rider_7646	Calgary	Boston	cash
29.99	Driver_862	Rider_2993	Los Angeles	Toronto	cash
29.99	Driver_1000	Rider_8405	Chicago	San Francisco	paypal
29.99	Driver_52	Rider_9784	Vancouver	Ottawa	card
29.99	Driver_1574	Rider_9364	Boston	Toronto	paypal
29.99	Driver_1832	Rider_7213	Ottawa	New York	voucher

Table 1. Result of the Top 10 longest rides by distance

Finding: The analysis reveals the maximum ride distances, associated driver-rider pairs, and payment methods used for premium long-distance services.

2. Rider Retention from 2021 – 2024

Objective: To find how many riders who signed up in 2021 still took rides in 2024.

```

with riders_2021 as(
    select
        rider_id
    from
        riders
    where
        extract(year from signup_date) = 2021
),
active_in_2024 AS (
    select distinct
        t1.rider_id
    from
        rides t1
    join
        payments t2 on t1.ride_id = t2.ride_id
    where
        t2.amount > 0
        and extract(year from t1.request_time) = 2024
)

select
    count(t1.rider_id) as riders_signup_2021_active_2024
from
    riders_2021 t1
join
    active_in_2024 t2 on t1.rider_id = t2.rider_id

```

Fig 2.

riders_signup_2021_active_2024
1815

Table 2. Result of the total active riders that signed up in 2021

Finding: A total of 1,815 riders who signed up in 2021, still took rides in 2024.

3. Quarterly Revenue Growth (YOY)

Objective: To compare quarterly revenue between 2021, 2022, 2023, and 2024. Also find what quarter had the biggest YoY growth.

```

with quarterly_revenue as (
  select
    extract(year from t1.request_time) as ride_year,
    extract(QUARTER from t1.request_time) as ride_quarter,
    sum(t2.amount) as total_revenue
  from
    rides t1
  join
    payments t2 on t1.ride_id = t2.ride_id
  where
    t2.amount > 0
  group by
    1, 2
),
yoy_growth as (
  select
    curr.ride_year,
    curr.ride_quarter,
    curr.total_revenue as current_revenue,
    prev.total_revenue as previous_revenue,
    ((curr.total_revenue - prev.total_revenue) / prev.total_revenue) * 100
    AS growth_percent
  from
    quarterly_revenue curr
  join
    quarterly_revenue prev
    on curr.ride_quarter = prev.ride_quarter
    and curr.ride_year = prev.ride_year + 1
)
select
  ride_year,|
  ride_quarter,
  growth_percent
from
  yoy_growth
order by
  growth_percent desc;

```

Fig 3.

ride_year	ride_quarter	growth_percent
2022	2	200.7956495
2024	4	3.903846117
2024	3	3.36212723
2022	4	2.772021819
2023	2	0.544210802
2022	3	0.057064034
2023	1	-1.932760595
2024	2	-4.79901803
2024	1	-5.220395644
2023	4	-5.415536037
2023	3	-5.688304388

Table 3. Result for quarterly revenue

Finding: The biggest YoY growth was 200.80% in Q2 2022, also the data show a negative growth percent in 2023 and early 2024.

4. Top 5 Drivers by Average Monthly Rides

Objective: To calculate each driver's average monthly rides since signup and who the top 5 drivers with the highest consistency (most rides per active month) are.

```
with driver_ride_summary as(
    select
        t1.driver_id,
        t2.name as driver_name,
        t2.signup_date,
        count(t1.ride_id) as total_rides,
        max (t1.request_time) as last_ride_time
    from
        rides t1
    join
        drivers t2 on t1.driver_id = t2.driver_id
    join
        payments t3 on t1.ride_id = t3.ride_id
    where
        t3.amount > 0
    group by
        t1.driver_id, t2.name, t2.signup_date
),
driver_consistency as (
    select
        driver_name,
        total_rides,
        (extract(year from last_ride_time) - extract(year from signup_date)) * 12 +
        (extract(month from last_ride_time) - extract(month from signup_date)) + 1
        as active_months,
        (total_rides * 1.0 / (
            (extract(year from last_ride_time) - extract(year from signup_date)) * 12 +
            (extract(month from last_ride_time) - extract(month from signup_date))+1
        )) as average_monthly_rides
    from
        driver_ride_summary
    where
        (extract (year from last_ride_time) - extract(year from signup_date)) * 12 +
        (extract(month from last_ride_time) - extract(month from signup_date))+ 1 > 0
)
select
    driver_name,
    total_rides,
    active_months,
    round(average_monthly_rides, 2) as avg_monthly_rides_per_active_month
from
    driver_consistency
order by
    average_monthly_rides desc
limit 5;
```

Fig 4.

driver_name	total_rides	active_months	avg_monthly_rides_per_active_month
Driver_1722	13	1	13
Driver_1690	18	3	6
Driver_1876	17	3	5.67
Driver_800	17	4	4.25
Driver_1158	18	5	3.6

Table 4. Result of the Top 5 consistent drivers.

Finding: Driver_1722 is the most consistent driver, averaging 13 completed rides per active month.

5. City with the Highest Cancellation Rate

Objective: To calculate the cancellation rate per city and identify which city had the highest cancellation rate.

```

with city_ride_metrics as (
    select
        pickup_city,
        count(ride_id) as total_rides_attempted,
        sum(case when status = 'cancelled' then 1 else 0 end)
            as total_cancelled_rides
    from
        rides
    where
        pickup_city is not null
    group by
        pickup_city
)

select
    pickup_city,
    total_rides_attempted,
    total_cancelled_rides,
    (total_cancelled_rides * 100.0 / total_rides_attempted)
        as cancellation_rate_percent
from
    city_ride_metrics
where
    total_rides_attempted > 0
order by
    cancellation_rate_percent desc;

```

Fig 5.

pickup_city	total_rides_attempted	total_cancelled_rides	cancellation_rate_percent
Chicago	4458	859	19.26873037
Toronto	4464	849	19.0188172
Calgary	4460	825	18.49775785
San Francisco	4430	814	18.37471783
Montreal	4377	797	18.20881883
New York	4361	787	18.04631965
Los Angeles	4405	793	18.00227015
Vancouver	4329	773	17.85631786
Ottawa	4536	805	17.74691358
Boston	4446	787	17.70130454

Table 5. Result of the cancellation rate.

Finding: Chicago had the highest operational challenge with a cancellation rate of 19.27%.

6. Riders with No Cash Payment

Objective: To identify riders who have taken more than 10 rides but never paid with cash.

```

with rider_payment_summary as (
    select
        t1.rider_id,
        t2.name as rider_name,
        count(t1.ride_id) as total_completed_rides,
        sum(case when t3.method = 'cash' then 1 else 0 end) as cash_payments_count
    from
        rides t1
    join
        riders t2 on t1.rider_id = t2.rider_id
    join
        payments t3 on t1.ride_id = t3.ride_id
    where
        t3.amount > 0
    group by
        t1.rider_id, t2.name
)

select
    rider_name,
    total_completed_rides,
    cash_payments_count
from
    rider_payment_summary
where
    total_completed_rides > 10
    and cash_payments_count = 0
order by
    total_completed_rides desc;

```

Fig 6.

rider_name	total_completed_rides	cash_payments_count
Rider_7823	12	0

Table 6. Result of riders who have never been paid with cash.

Finding: Rider_7823 was the only ride who had never been paid with cash (a predominant digital payment user) and he had taken a total of 12 rides.

7. Top 3 Drivers in Each City by Total Revenue

Objective: To find the top 3 drivers in each city by total revenue earned between June 2021 and Dec 2024, and if a driver has multiple cities, count revenue where they picked up passengers in that city.

```
with city_driver_revenue as (
    select
        r.pickup_city,
        d.name as driver_name,
        sum(p.amount) as total_revenue,
        rank() over (partition by r.pickup_city order by sum(p.amount) desc)
            as city_rank
    from
        rides r
    join
        drivers d on r.driver_id = d.driver_id
    join
        payments p on r.ride_id = p.ride_id
    where
        p.amount > 0
    group by
        1, 2
)

select
    pickup_city,
    driver_name,
    total_revenue
from
    city_driver_revenue
where
    city_rank <= 3
order by
    pickup_city, total_revenue desc;
```

Fig 7.

pickup_city	driver_name	total_revenue
Boston	Driver_1176	448.4
Boston	Driver_286	326.58
Boston	Driver_1141	315.88
Calgary	Driver_1980	476.91
Calgary	Driver_1059	346.86
Calgary	Driver_404	338.8
Chicago	Driver_413	449.45

Chicago	Driver_1410	421.9
Chicago	Driver_1941	331.53
Los Angeles	Driver_761	433.12
Los Angeles	Driver_448	373.29
Los Angeles	Driver_287	334.24
Montreal	Driver_163	377.87
Montreal	Driver_1328	341.06
Montreal	Driver_541	337
New York	Driver_681	338.41
New York	Driver_1708	318.65
New York	Driver_1910	303.01
Ottawa	Driver_418	358.1
Ottawa	Driver_76	353.81
Ottawa	Driver_645	309.7
San Francisco	Driver_286	354.75
San Francisco	Driver_13	352.62
San Francisco	Driver_1626	352.13
Toronto	Driver_988	380.56
Toronto	Driver_372	363.52
Toronto	Driver_383	322.63
Vancouver	Driver_1924	365.35
Vancouver	Driver_508	358.06
Vancouver	Driver_578	329.28

Table 7. Result of the Top 3 drivers in each city.

Finding: The driver that produced the most revenue across all cities is Driver_1980 (Calgary) with 476.91 in total earnings.

8. Top 10 Drivers Qualified for Bonus

1. **Objective:** To find the top 10 drivers that are qualified to receive bonuses using the criteria below;
 - at least 30 rides completed,
 - an average rating ≥ 4.5 , and
 - a cancellation rate under 5%.

```

with driver_performance_metrics as (
    select
        r.driver_id,
        d.name as driver_name,
        d.rating as driver_static_rating,
        sum(case when p.amount > 0 then 1 else 0 end) as completed_rides,
        count(r.ride_id) as total_rides_attempted,
        sum(case when r.status = 'cancelled' then 1 else 0 end)
            as total_cancelled_rides
    from
        rides r
    join
        drivers d on r.driver_id = d.driver_id
    left join
        payments p on r.ride_id = p.ride_id
    group by
        1, 2, 3
)

select
    driver_name,
    completed_rides,
    driver_static_rating,
    round((total_cancelled_rides * 100.0 / total_rides_attempted), 2)
        as cancellation_rate_percent
from
    driver_performance_metrics
where
    completed_rides >= 30
    and driver_static_rating >= 4.5
    and (total_cancelled_rides * 100.0 / total_rides_attempted) < 5.0
order by
    completed_rides desc
limit 10;

```

Fig 8.

driver_name	completed_rides	driver_static_rating	cancellation_rate_percent
Driver_1005	31	4.8	0
Driver_1181	31	4.6	3.13

Table 8. Result of the top 10 drivers that qualify for bonus

Finding: Only 2 drivers meet all three bonus criteria given by the management.

Conclusion

The analysis comprehensively evaluated HNG Ride's operational and financial performance between **June 2021 and December 2024**, validating key metrics across driver activity, rider retention, and revenue generation. HNG Ride maintained strong rider loyalty and consistent driver performance, but experienced a slowdown in revenue growth. The 19.27% cancellation rate in Chicago highlights a critical operational challenge requiring immediate intervention. Additionally, the current

bonus criteria, though promoting excellence, are overly restrictive, with only two qualifying drivers.